Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/18/2024
PI Name:	Ronca, April Elizabeth Ph.D.		
Project Title:	Rodent Research Standard Housing Mission: Multidisciplinary Approach to Understanding Spaceflight Responses		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Immunology		
Space Biology Special Category:	None		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:	November 2019: PI is located at NASA Ames Research Center and remains affiliated with Wake Forest University School of Medicine		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/01/2022	**End Date:**	04/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Task Description:	RR-26 is a NASA Space Biology (SB) Program funded study for the 2nd Rodent Habitat Flight Validation, following the addition of the Nest Box (NB) as the current solution to meet OCHMO-550 requirements and to provide adequate bedding substrate and/or structures for resting and sleeping of animals per The Guide to the Care and Use of Laboratory Animals (The Guide). The Flight Institutional Animal Care and Use Committee (FIACUC) and the science community share a common goal to advance the utility of the Rodent Research (RR) Habitat for conducting high quality, rodent spaceflight experiments. The NASA Chief Veterinarian and FIACUC are concerned that the RR Habitat without a Nest Box (NB), fails to meet animals' needs and the regulatory requirements to provide a dedicated structure that allows mice to sleep, nest, and thermoregulate. On the other hand, growing concerns have been raised in the RR flight investigator community (based on results obtained from some missions since 2017) that using the NB significantly alters the environment and may interfere with expected physiological responses to spaceflight that were identified previously and expected during long duration missions, thereby diminishing the scientific value of the RR animal model used to study biological effects of spaceflight.This mission is directed research by Space Biology and the plan was developed following extensive consultation and collaboration with the NASA Chief Veterinarian, Flight IACUC, the RR Science Working Group (SWG), welfare, and other science experts. This mission will provide reference data on animals' physiological systems and behavior within the Rodent Habitat to inform the NB's role in 1) supporting rodent health and welfare in space, and 2) to inform how the NB affects key tissues and science disciplines in both the space and ground environment. To accomplish mission objectives, adult wild-type female mice will be flown to the International Space Station (ISS) using RR Hardware. Groups of mice provided with the NB will be compared to those without the NB in the Rodent Habitat in both Flight and Ground. Comparison of Flight groups (FLT) to their respective Habitat Ground Control (HGC) groups will reveal if and how the NB affects major physiological systems relevant to the discipline of Space Biology.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal. By performing directed research, Space Biology (SB) plans to conduct targeted ground and flight studies to obtain the required data and to enable critical, scientifically rigorous analyses of comparing the effects with and without the Nest Box on rodents during spaceflight.
Research Impact/Earth Benefits:	The Rodent Research Standard Housing Mission (RR-26) is a critical multi-disciplinary experiment designed to assess and improve translation of NASA rodent spaceflight studies with the goal of informing crew health on long duration missions. This will be accomplished b providing evidence to: (1) help define the role of a physical structure, or Nestbox, in supporting aspects of rodent. health and welfare in space, and determine whether provision of the Nestbox limits the study of certain scientific disciplines.
Task Progress:	DECEMBER 2024 UPDATE: Per NASA Ames Research Center (ARC), Dr. April Ronca has retired from NASA, and is no longer the Principal Investigator of this project. The project continues with the same name under a new PI, Yasaman Shirazi-Fard, Ph.D. (Ed., 12/23/24). Preparations for the RR-26 mission have continued during this second grant year, and major mission milestones were achieved. An RR-26 Technical Interchange Meeting (TIM) was held at NASA Ames Research Center in June 2023 to address specific critical topics related to RR-26 science development, operations, and implementation. A major augmentation of the RR-26 mission involves upgrading existing analog cameras to digital technology, and continuous rather than discrete video sampling of rodent behavior for analysis. In the second year, focus has been on coordinating science requirements with video engineering capabilities. To support inflight data collection, two studies were conducted to establish the feasibility of utilizing the new Habitat digital cameras to identify individual mice via fur dye markings (Aim 1), continuous body temperature acquired using implanted dataloggers (Aim 2), and assessment of the circadian timing system using video and body temperature data to assess parallel changes across light and dark cycle phases (Aim 3). Due to the current technical inability to measure mouse sleep on the International Space Station/ISS (Aim 4), we sought proof-of-concept that sleep-like behavior can be inferred from these circadian-sensitive measures. Aims 5-9, critical for addressing the role(s) of the Nest Box (NB) in brain, bone, muscle, immune, and cardiovascular responses to spaceflight, require little or no preliminary testing owing to numerous past spaceflight studies. For these reasons, only Science Tests supporting RR-26 Aims 1-4 are described in the Test Plan Document No. 8020, RR26 Science Tests to Validate Fur Bleach Patterns and Datalogger Implants in Mice. RESULTS ST-1 and -3 Main Findings Supported Aims 1-4. Objectives were successfully met as described here: ST-1 • The Launch Simulation Test successfully validated that datalogger implantation did not damage or elicit concerning pathological responses in the mouse abdominal cavity (NASA Attending Vet Necropsy report). Continuous acquisition of datalogger temperature measures was reliable, and unaffected by launch forces. • Datalogger use in the Habitat and under launch simulation conditions has been successfully vetted and established with zero concerns. ST-3 • Individual identification from fur dye patterns in combination with ear notches is attainable within the RR-26 habitat with a full complement of mice (5 per side). In this study, fur dyes were readable for more than two weeks. • Optimization tests conducted by Dr. Alberts at Indiana University have since extended fur durability duration to the 30-day mission length. • The video system in its current configuration is adequate. It will provide "first ever" views and data, and thus can make substantial contributions to RR-26 science and to Space Biology more generally. However, there is much room for improvement, and we strongly support NASA in further developing this system using available and affordable technologies. • Preliminary analysis of huddling behavior relative to body temperature suggests the possibility of identifying meaningful relations between these measures; however, analysis was limited by short duration video samples. CONCLUSION The data collected from the implanted dataloggers will provide continuous measurement of the body temperature of

	each animal throughout the RR-26 mission, beginning with preflight control measurements. These data are integral for the peer-reviewed outcomes of Specific Aims 2 & 3. For these Aims, we will utilize the body temperature data provided by the data loggers to determine how the Circadian Timing System responds and adapts to the spaceflight environment. Moreover, the data will be used to understand metabolism and temperature regulation during the mission. These data will also be extremely valuable to other investigators. Data from the loggers will provide a direct physiological indication of the overall health and well-being of the subjects, as well as each animal's adaptation pattern to microgravity, over the course of the mission. Given robust segments of continuous video acquired simultaneously from each habitat's camera set, changes in body temperature data can also be analyzed in combination with the recorded behavioral observations to bolster information on the rest-activity patterns of each animal as surrogate measures of their sleep wake regulation. This is especially important as we cannot, currently, record EEG to measure sleep/wake directly. Finally, we will be able to, for the first time, make direct correlations between a physiological variable and an animal's individual and group behaviors.
Bibliography Type:	Description: (Last Updated: 05/15/2025)